

Parkigait

On-device gait screening for Parkinson's — research prototype, NOT a medical device

What it is

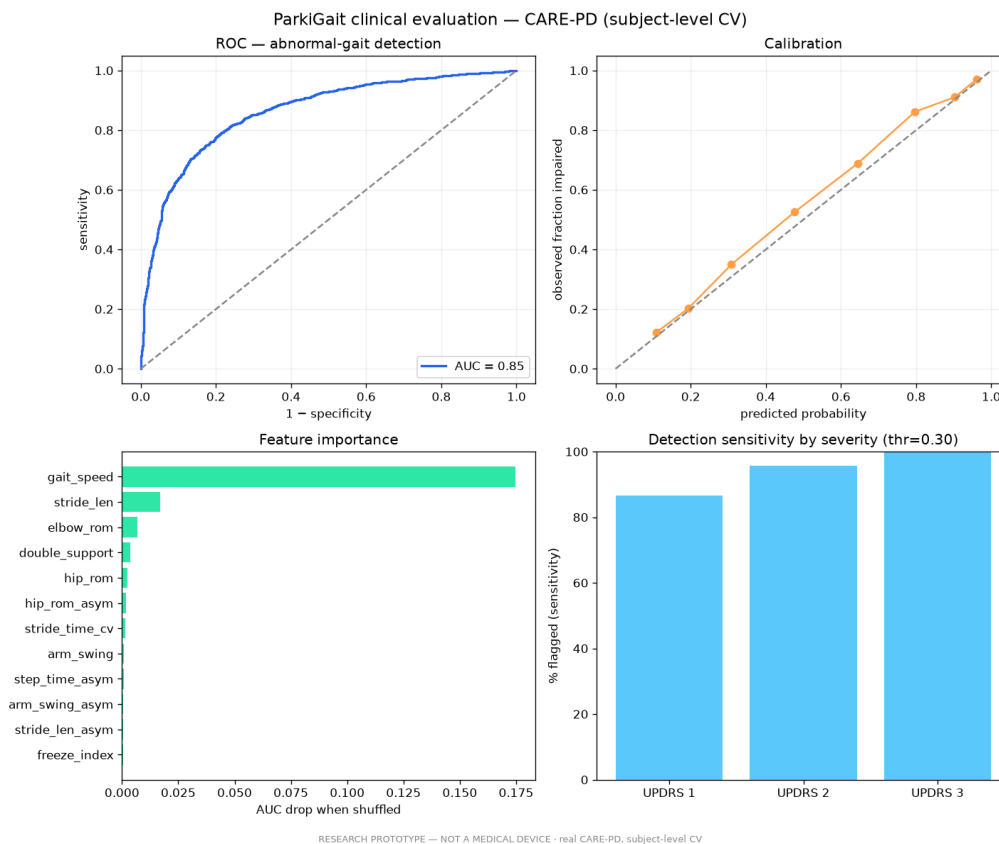
A tool that takes a short walking video, extracts a skeleton on the device, computes clinically grounded gait features (walking speed, cadence, stride length, joint range of motion, arm swing, trunk flexion, step to step variability, freezing index), and flags possible parkinsonian gait for a clinician to review. It runs fully on a phone or laptop with no cloud, so patient video never leaves the device.

Results on real clinical data

Trained and evaluated on the public CARE-PD dataset (110 patients with clinician MDS-UPDRS gait scores, ~2953 walks). Strict subject level cross validation, 95% confidence intervals by patient level bootstrap.

Task	Result
Detect impaired gait (screening) — AUC	0.86 to 0.87 (95% CI 0.82 to 0.90), permutation $p < 0.005$
External validation (a hospital never seen in training)	pooled AUC 0.77 (0.81 to 0.86 at the larger sites)
Severity staging (UPDRS-gait 0 to 3)	quadratic weighted kappa 0.62, 95% within one level
Screening operating point	sensitivity 0.90, specificity 0.59, PPV 0.75, NPV 0.81
Test-retest reliability (repeated walks)	ICC 0.63, calibrated probability (Brier 0.15)
Speed / memory / privacy	~27 ms per frame, under 400 MB, on-device (no cloud)

The strongest single feature is walking speed, followed by stride length and joint range of motion, the same signs a clinician looks for. Leakage was independently audited and a permutation test confirms the result is not chance.



What it is not

Not a diagnosis, not a replacement for a clinician or the standard exam, and not ready for any patient care decision. It is a research and screening aid.

What I am seeking

Feedback from a clinician on where this could genuinely help, and if you are open to it a small research collaboration under ethics approval to compare the tool's flag to your own assessment on anonymized gait videos.

Parv Mehndiratta | open source code and full honest documentation: github.com/0103-parv/parkinsons-detection | joints are a canonical-skeleton approximation; validated on one public dataset; see the model card and clinical evaluation in the repo.